

Subject Name: Python for Data Science

Sr. No	List of Experiment
Unit 1 – Introduction, Data Types, Operators, Loops & Functions	
1	Write Python programs to demonstrate variables, data types (int, float, str, bool) and all operators (Arithmetic, Assignment, Comparison, Logical, Membership, Identity, Bitwise).
2	Demonstrate creation and manipulation of list, tuple, set, dictionary and string with their built-in methods.
3	Implement programs using if, if-else, elif, while loop, for loop, nested loops, and loop control statements (break, continue, pass).
4	Write Python programs using user-defined functions with different argument types and demonstrate use of built-in modules (math, random, statistics).
Unit 2 – File Handling, Exception Handling & MySQL	
5	Perform read, write and append operations on text files, CSV files, and binary files using Python.
6	Demonstrate exception handling using try, except, else, finally blocks and raise statement with custom exceptions.
7	Connect Python with MySQL database and perform Create, Read, Update and Delete (CRUD) operations.
Unit 3 – Data Science & Descriptive Statistics	
8	Calculate and interpret descriptive statistics: mean, median, mode, variance, standard deviation, range, quartiles and IQR on a given dataset.
9	Demonstrate the data science life cycle and perform Univariate, Bivariate and Multivariate analysis on a real-world dataset.
Unit 4 – Exploring Python Libraries (NumPy, Pandas, SciPy, Scikit-learn, BeautifulSoup)	
10	Demonstrate NumPy array creation, indexing, slicing, reshaping, broadcasting and mathematical operations on multidimensional arrays.
11	Use Pandas to create Series and DataFrames, perform data selection, filtering, grouping and aggregation operations.
12	Use SciPy's stats module to compute statistical measures and perform statistical tests on datasets.
13	Extract and parse data from an HTML document or web page using BeautifulSoup library.
14	Demonstrate basic machine learning workflow using Scikitlearn: data loading, splitting, model training and evaluation.
Unit 5 – Probabilistic and Inferential Statistics	

15	Compute and visualize Binomial and Poisson probability distributions using SciPy and Matplotlib.
16	Plot and analyze Normal and Exponential continuous probability distributions and demonstrate the Central Limit Theorem.
17	Perform hypothesis testing using one sample t-test and interpret results based on p-value and significance level.
Unit 6 – Data Preparation & Preprocessing	
18	Perform data cleaning operations: detect and handle missing values, remove duplicate records, and treat outliers in a dataset.
19	Apply data preprocessing techniques: slicing, filtering, sorting, concatenating, aggregating and normalizing data using Pandas.
Unit 7 – Data Visualization	
20	Create various types of plots using Matplotlib: line plot, bar chart, histogram, pie chart, scatter plot using subplots and figure properties.